

Soutenance de thèse

Apprentissage actif non supervisé

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Sous la direction de Nicolas Verzelen[†] et Alexandra Carpentier^{*}

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Introduction to unsupervised active learning

Focus on Active Clustering

Focus on Condorcet Winner Identification

Conclusion and Perspectives

Introduction to unsupervised active learning

Unsupervised Active Learning

Learning: use data to solve a task

Active: choose data sequentially and adaptively

Unsupervised: recover hidden structure without labels

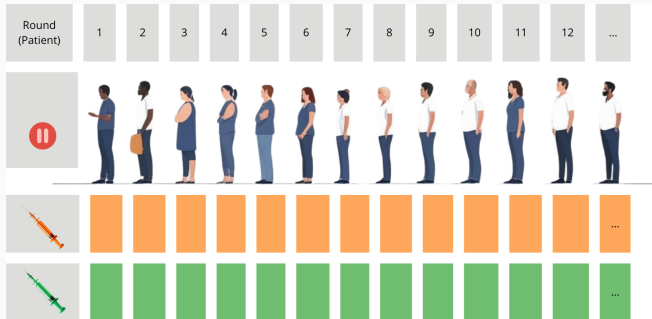
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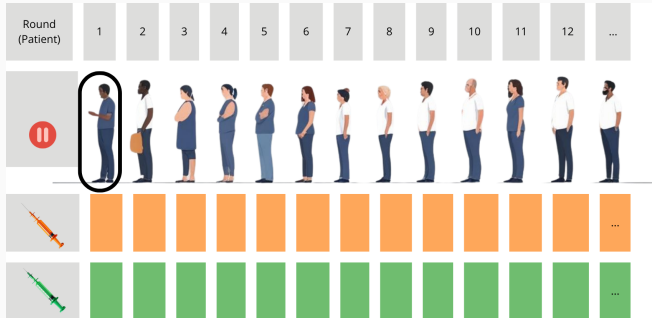
**Pure Exploration Problems
in a bandit environment
with hidden structure**

Bandit model: Sequential and Adaptive learning (Thompson, 1933)



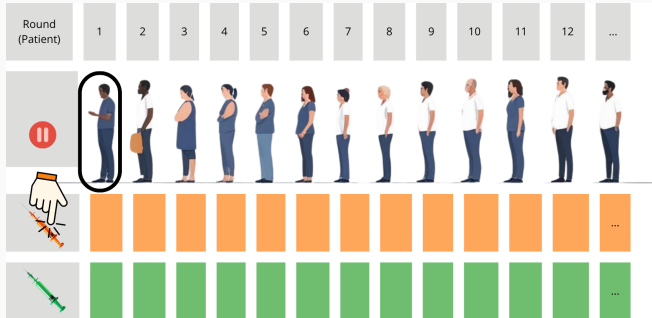
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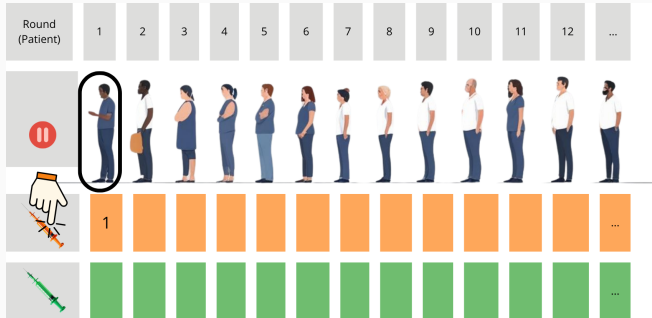
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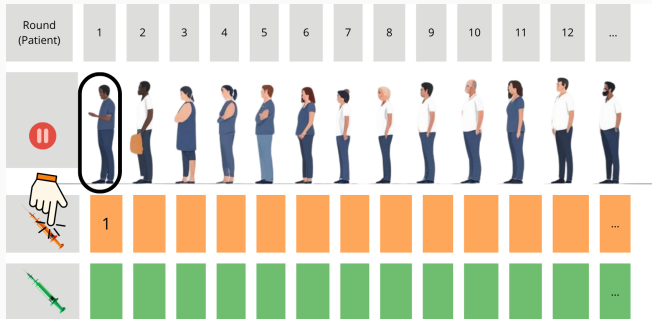
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- Pure exploration (Even-Dar, Mannor, and Mansour, 2002)
 - ↪ recover a hidden structure/characteristic (e.g., the best drug)
 - ↪ minimize sampling budget (e.g., number of patients)



Pure exploration in bandit problems with hidden structure

- **Feature-matrix** M of shape $n \times d$ (n items and d features)

					
	0	0	0	0	0
	0	0.5	0.05	0	0.1
	0	0.5	0.05	0	0.1
	0	0	0	0	0

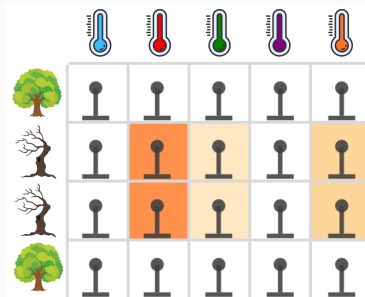
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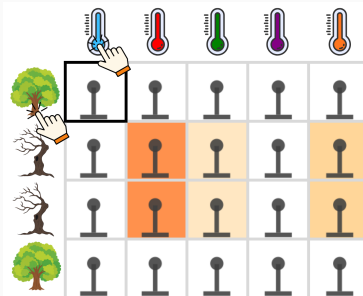
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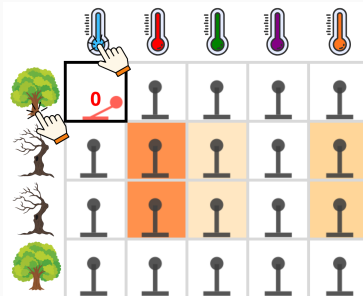
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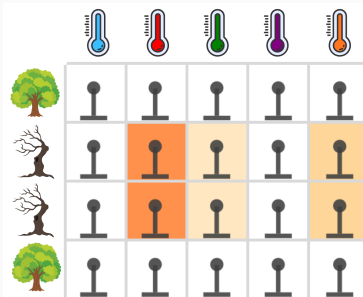
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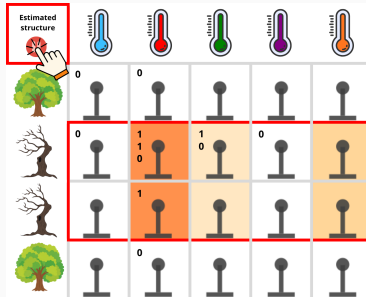
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Fixed confidence setting

Given a confidence level δ , recover $\mathcal{C}^*(M)$ with probability at least $1 - \delta$, while minimizing the sampling budget.

- such an algorithm is called δ -correct
- its performance is measured by the number of samples it uses, i.e., the **sampling budget** N_δ

Overview of models

1. Parametric clustering with bandit (Thuot*, Carpentier, Giraud, and Verzelen, 2025)
2. Non-parametric clustering with bandit (Thuot*, Vogt*, Ghoshdastidar, and Verzelen, 2026)
3. Clustering with adaptive feature selection (Graf*, Thuot*, and Verzelen, 2025)
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5. Change-point detection (Graf* and Thuot*, 2026)

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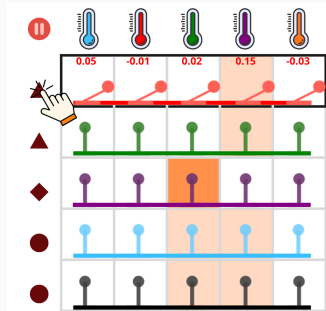
Setting

- goal: **clustering** of rows
- observation: **full-row**
- **parametric** assumption
($d < +\infty$ + sub-Gaussian noise)

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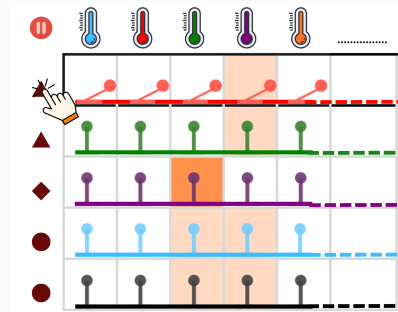
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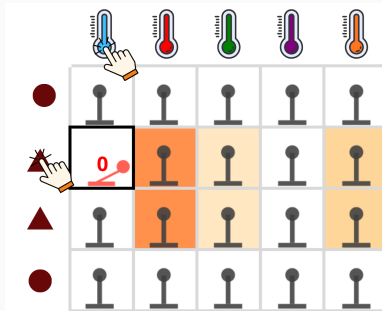
Setting

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- non-parametric assumption ($d \leq \infty$ + any noise)

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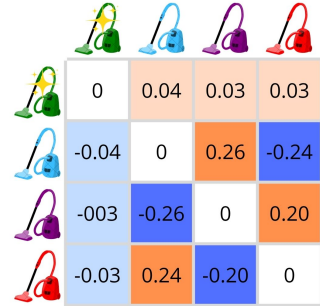
Setting

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- observation: single-feature
- parametric assumption ($d < +\infty$ + sub-Gaussian noise)

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	0	0.04	0.03	0.03
	-0.04	0	0.26	-0.24
	-0.03	-0.26	0	0.20
	-0.03	0.24	-0.20	0

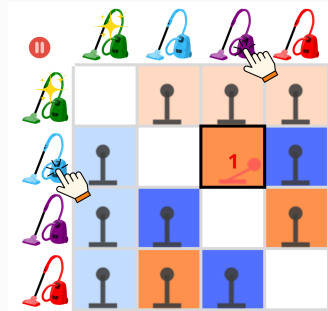
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- observation: **pairwise comparisons**
(random duel I_t versus J_t)
- **Bernoulli observation**

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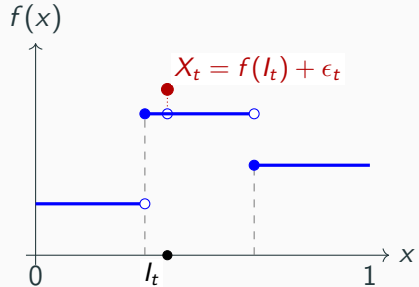
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Setting

- goal: recover the **change points locations** (at precision η)
- observation: noisy **evaluations** of f
- action set: **continuous** $[0, 1]$

Problem positioning

- Unsupervised tasks

Examples: clustering, kernel clustering, signal detection, and change-point detection (Kaufman et al., 2009; Dhillon et al., 2004; Basseville et al., 1993)

High-dimensional problem (Rigollet et al., 2023; Giraud, 2021)

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- Pure exploration for bandits

Either 1-D (BAI, thresholding, signal detection)

(Lattimore et al., 2020; Bubeck et al., 2012; Castro, 2014)

Or Asymptotic analysis ($\delta \rightarrow 0$)

(Kaufmann et al., 2016; Karnin, 2016; Yang et al., 2024; Lazzaro et al., 2025)

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Or **Asymptotic analysis** ($\delta \rightarrow 0$)

(Kaufmann et al., 2016; Karnin, 2016; Yang et al., 2024; Lazzaro et al., 2025)

- **Dueling bandits**

(Bengs et al., 2021; Haddenhorst et al., 2021; Maiti et al., 2024; Karnin, 2016)

Main questions

- **Q1.** How can we recover the hidden structure efficiently?

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- **Q2.** What is the minimal sampling budget required?
- **Q3.** What are the advantages of a non-asymptotic analysis?
- **Q4.** What are the advantages of bandit learning over classical (batch) learning?

Focus on Active Clustering

Introduction to unsupervised active learning

Focus on Active Clustering

- Setting

- Lower bound

- Upper Bound and algorithmic ideas

- Take Home Message

Focus on Condorcet Winner Identification

Conclusion and Perspectives

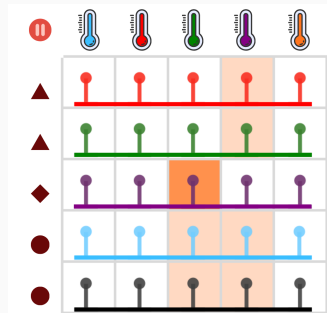
Active Clustering Model (Yang, Zhong, and Tan, 2024)

- each item $a \in [n] \leftrightarrow \mathcal{N}(\mu_a, \sigma^2 I_d)$, with $\mu_a \in \mathbb{R}^d$ unknown
- **Objective:** recover the **hidden partition** G^* of $[n]$ into K **groups**, such that a, b are in the same group $\Leftrightarrow \mu_a = \mu_b$.

					
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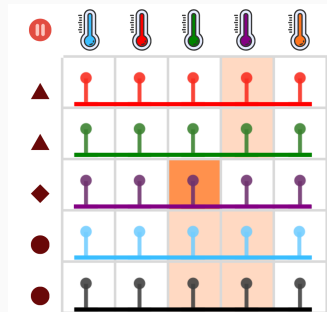
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- At round t ,
 1. **Choose** one item $I_t \in [n]$ to sample
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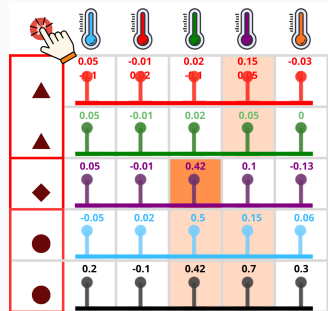


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- Fixed confidence:** recover G^* with probability at least $1 - \delta$, while minimizing the number of samples $\rightarrow \delta$ -correct

Model parameters

Parameters

- number of items n
- Dimension of the feature vectors d
- Number of clusters K (assumed to be known)
- Confidence parameter: δ
- Noise variance: σ^2
- Minimal gap: $\Delta_* = \min_{\mu_a \neq \mu_b} \|\mu_a - \mu_b\| > 0$

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Sampling complexity

$$T^*(\delta) := n + \frac{\sigma^2}{\Delta_*^2} n \log \left(\frac{n}{\delta} \right) + \frac{\sigma^2}{\Delta_*^2} \sqrt{dKn \log \left(\frac{n}{\delta} \right)}$$

The problem was introduced in (Yang, Zhong, and Tan, 2024), they provide BOC algorithm such that

1. Track-and-Stop method (Garivier and Kaufmann, 2016)
2. For balanced and equidistant groups, budget is equivalent to

$$\mathbb{E}[N_\delta] \underset{\delta \rightarrow 0}{\sim} 2 \frac{\sigma^2}{\Delta_*^2} (n + K) \log(1/\delta)$$

3. Asymptotic optimality when $\delta \rightarrow 0$ with n, K, d, Δ_* fixed (exact constant)
4. But hides high-dimensional effects

Benchmark: batch learning

Uniform sampling strategy (US):

1. sample each item the same number of times $\frac{T_{\text{oracle}}}{n}$
2. use (batch) clustering on the empirical means

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- Information limit:

$$T_{\text{oracle}} \gtrsim \frac{\sigma^2}{\Delta^2} \left[n(\log(n)) + \sqrt{dKn(\log(n))} \right] \simeq T^*(1)$$

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- State-of-the-art polynomial-time algorithms:

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(Royer, 2017; Giraud and Verzelen, 2019; Vempala and Wang, 2004)

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Computational gap in batch clustering (Even, Giraud, and Verzelen, 2024)

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Theorem (Thuot*, Carpentier, Giraud, and Verzelen, 2025)

Let $\mathcal{A}$ be a δ -correct algorithm. There exists an environment ν with minimal gap Δ_* , balanced groups, such that

$$\mathbb{E}_{\mathcal{A}, \nu}[N_\delta] \geq cT^*(\delta)$$

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Upper bound: Active Clustering with Bandits (ACB)

ACB[†] Algorithm

1: **while** true **do**

2: **Doubling step:** $T \leftarrow 2^r$

12: **end while**

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2: **Doubling step:** $T \leftarrow 2^r$

3: **Representative search:** find one item from each group ($\hat{S}$)

budget: 2^r samples

12: **end while**

Upper bound: Active Clustering with Bandits (ACB)

ACB† Algorithm

```
1: while true do
```

2: **Doubling step:** $T \leftarrow 2^r$

3: **Representative search:** find one item from each group ($\hat{S}$)

budget: 2^r samples

4: **Center estimation:** estimate unknown centers from items in $\hat{S}$

budget: T/K samples per item in $\hat{S}$

```
12: end while
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Upper bound: Active Clustering with Bandits (ACB)

ACB† Algorithm

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1: while true do
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2: **Doubling step:** $T \leftarrow 2^r$

3: **Representative search:** find one item from each group ($\hat{S}$)

budget: 2^r samples

4: **Center estimation:** estimate unknown centers from items in $\hat{S}$

budget: T/K samples per item in $\hat{S}$

5: **Classification:** assign each item to its closest estimated center

budget: T/n samples per item

```
12: end while
```

Upper bound: Active Clustering with Bandits (ACB)

ACB† Algorithm

```

1: while true do
2:   Doubling step:  $T \leftarrow 2^r$ 
3:   Representative search: find one item from each group ( $\hat{S}$ )
                                     budget:  $2^r$  samples
4:   Center estimation: estimate unknown centers from items in  $\hat{S}$ 
                                     budget:  $T/K$  samples per item in  $\hat{S}$ 
5:   Classification: assign each item to its closest estimated center
                                     budget:  $T/n$  samples per item
6:   Verification: test candidate partition  $\hat{G}_T$ 
7:   if  $\hat{G}_T$  is certified then
8:     return  $\hat{G}_T$ 
9:   else
10:     $r \leftarrow r + 1$  and repeat
11:  end if
12: end while

```

Upper bound: main theorem

Theorem (Thuot*, Carpentier, Giraud, and Verzelen, 2025)

(i) $\text{ACB}_{\dagger}^{\nu}$ is δ -correct.

(ii) For $n \geq K \log(K)$, and balanced groups, then

$$\mathbb{E}_{\text{ACB}_{\dagger}^{\nu}}[T] \leq c T^*(\delta) = c \left(n + \frac{\sigma^2}{\Delta^2} n \log(n/\delta) + \frac{\sigma^2}{\Delta^2} \sqrt{dKn \log(n/\delta)} \right)$$

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- Achieved by $\text{ACB}^\dagger$ (polynomial time)

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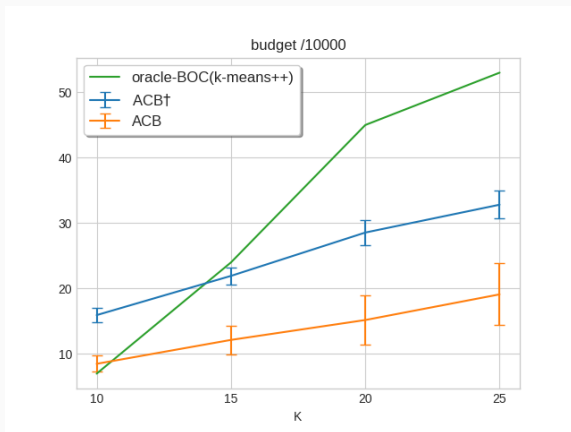
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Main message

No information-computation gap in active clustering

Figure 1: Comparison of sampling budget for $ACB^\dagger$ versus uniform sampling (oracle-BOC)



Take Home Message

- **Q1.** How can we recover the hidden structure efficiently?

ACB Algorithm : decompose the task into **center estimation**, **reconstruction**, and **certification**

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Estimation / Reconstruction trade-off

- **Q3.** What are the advantages of a non-asymptotic analysis?

Unveil the dependence in d , **high-dimensional bottleneck**

- **Q4.** What are the advantages of bandit learning over classical (batch) learning?

Better **sampling efficiency** and adaptivity

Better computational efficiency: **there is no computational gap**

Focus on Condorcet Winner Identification

Introduction to unsupervised active learning

Focus on Active Clustering

Focus on Condorcet Winner Identification

Setting

Lower bound

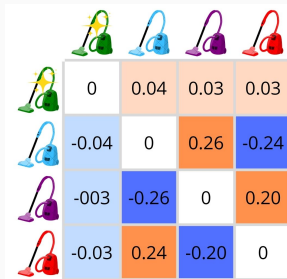
Upper bound and algorithmic ideas

Take-home message

Conclusion and Perspectives

Condorcet Winner Identification: problem setting

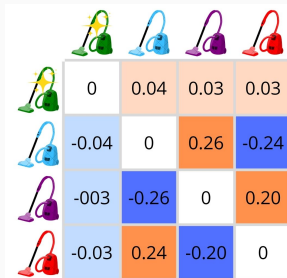
- K items
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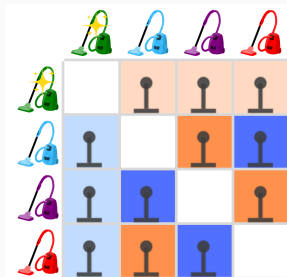
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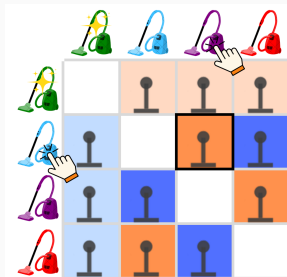
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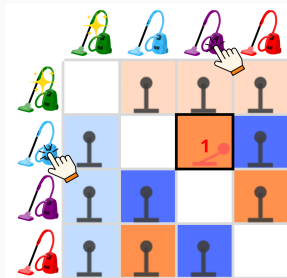
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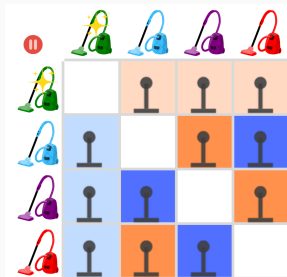
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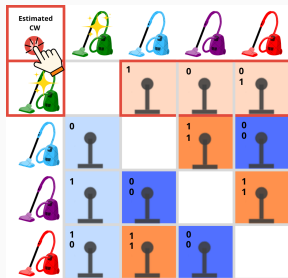
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Certification cost

Prove $\Delta_{i,j} < 0$ with confidence $1 - \delta$ costs $\sim \Delta_{i,j}^{-2} \log(1/\delta)$ samples

Baselines

Winner-row only (Maiti, Boczar, Jamieson, and Ratliff, 2024):

certify sub-optimality of arm i against the winner i^* :

$$\mathbb{E}[N_\delta] \lesssim H_{\text{cw}}(\delta) = \sum_{i \neq i^*} \frac{\log(1/\delta)}{\Delta_{i^*,i}^2}$$

Optimal asymptotic verification (Karnin, 2016):

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$$\mathbb{E}[N_\delta] \underset{\delta \rightarrow 0}{\sim} H_{\text{asyp}}(\delta) = \sum_{i \neq i^*} \min_{j: \Delta_{i,j} < 0} \frac{\log(1/\delta)}{\Delta_{i,j}^2}$$

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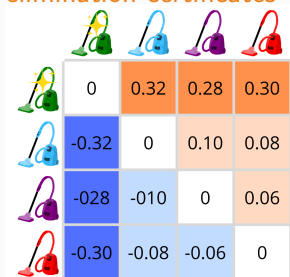
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Illustrations: easy vs. hard instances

Easy instance

i^* gives the strongest elimination certificates



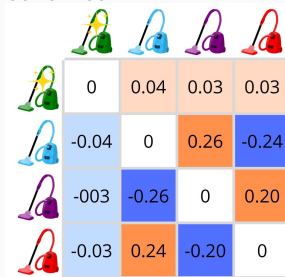
A 4x4 matrix illustrating an 'Easy instance'. The rows and columns are labeled with icons of four vacuum cleaners: green, blue, purple, and red. The diagonal elements are all 0. The off-diagonal elements are relatively large in magnitude, indicating strong elimination certificates. The matrix is symmetric.

				
	0	0.32	0.28	0.30
	-0.32	0	0.10	0.08
	-0.28	-0.10	0	0.06
	-0.30	-0.08	-0.06	0

- Holds in Strong Stochastic Transitivity (SST) settings
- $H_{\text{cw}}(\delta) = H_{\text{asympt}}(\delta)$ optimal (Maiti et al., 2024)

Hard instance

i^* almost even with at least one other item



A 4x4 matrix illustrating a 'Hard instance'. The rows and columns are labeled with icons of four vacuum cleaners: green, blue, purple, and red. The diagonal elements are all 0. The off-diagonal elements are small in magnitude, indicating weak elimination certificates. The matrix is symmetric.

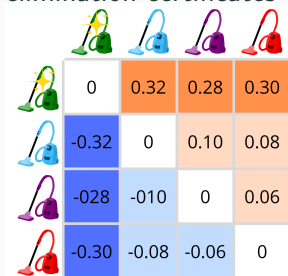
				
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	-0.04	0	0.26	-0.24
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- $H_{\text{cw}}(\delta)$ very large
- $H_{\text{asympt}}(\delta)$ optimal as $\delta \rightarrow 0$ (Karnin, 2016)

Illustrations: easy vs. hard instances

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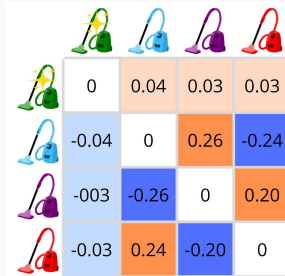
The table shows a 4x4 payoff matrix for an 'Easy instance'. The rows and columns are labeled with icons of vacuum cleaners: green, blue, purple, and red. The diagonal elements are all 0. The off-diagonal elements are positive values ranging from 0.06 to 0.32, indicating that the green vacuum cleaner (the optimal one) provides strong elimination certificates for all other items.

				
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The table shows a 4x4 payoff matrix for a 'Hard instance'. The rows and columns are labeled with icons of vacuum cleaners: green, blue, purple, and red. The diagonal elements are all 0. The off-diagonal elements are small values, both positive and negative, ranging from -0.26 to 0.26, indicating that the green vacuum cleaner is almost even with other items.

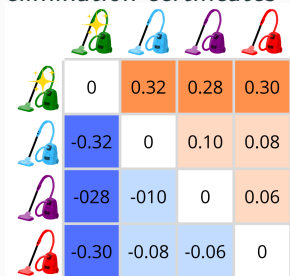
				
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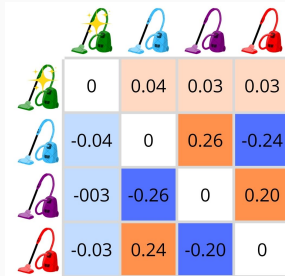
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The table shows a 4x4 payoff matrix for a 'Hard instance'. The rows and columns are labeled with icons of vacuum cleaners: green, blue, purple, and red. The diagonal elements are all 0. The off-diagonal elements are small and mixed in sign, indicating that the green vacuum cleaner is almost even with at least one other item.

				
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- $H_{\text{asympt}}(\delta)$ optimal as $\delta \rightarrow 0$ (Karnin, 2016)

For large K , $H_{\text{asympt}}(\delta)$ misses an intrinsic exploration cost.

Trade off exploration-certification

Notation: order the K_i negative gaps in row i as

$$\Delta_{i,(1)} \leq \dots \leq \Delta_{i,(K_i)} < 0$$

Fix $s_i \leq K_i$, there are s_i acceptable witnesses satisfying $\Delta_{i,j} \leq \Delta_{i,(s_i)}$.

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Search for one of the s'_i strongest opponents

Exploration

Finding one of the top- s_i strongest adversary costs

$$\frac{K}{s_i \Delta_{i,(s_i)}^2}.$$

Larger s_i makes the witness easier to find, but weaker to certify.

Certification

Once a witness is found, proving its sign costs

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Global trade-off

For $s = (s_i)_{i \neq i^*}$, the certification costs add over arms, whereas exploration is governed a non-trivial interplay with δ :

Exploration

Exploration over all arms costs

$$H_{\text{explore}}(s, \delta) = \sum_{i \neq i^*} \frac{K}{s_i \Delta_{i,(s_i)}^2} + \max_{i \neq i^*} \frac{K \log(1/\delta)}{s_i \Delta_{i,(s_i)}^2}.$$

Certification

Certification over all arms costs

$$H_{\text{certify}}(s, \delta) = \sum_{i \neq i^*} \frac{\log(1/\delta)}{\Delta_{i,(s_i)}^2}$$

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Theorem (local-minimax) (Saad*, Thuot*, and Verzelen, 2026)

Let $\mathcal{A}$ be a δ -correct algorithm. Then, one can construct $\tilde{\Delta}$ with the same sign structure as Δ , such that

$$\mathbb{P}_{\tilde{\Delta}, \mathcal{A}} \left(N_{\delta} \gtrsim \min_s (H_{\text{certify}}(s, \delta) + H_{\text{explore}}(s, \delta)) \right) \geq \delta.$$

Interpretation

The **exploration-certification trade-off** is intrinsic and unavoidable.

Fixed-budget subroutine: FB-CWI

At round r (Active set A_r)

1. $\forall i \in A_r$: compute the strong and weak opponent
2. Assign a score $S_r(i)$ using the computed statistics:
$$S_r(i) = \min\left\{\widehat{\Delta}_{i,i^{(s)}}, 0\right\} + \widehat{Q}_{[1/8, 1/4]}((\Delta_{i,j})_{j \in A_r}).$$
3. Rank the active arms by $S_r(\cdot)$ and discard the bottom $1/8$ fraction.

Strong opponent

Search over all $j \in [K] \setminus \{i\}$ with Bracketed Sequential Halving:

$$i^{(s)} = \widehat{\arg \min_{j \neq i} \Delta_{i,j}}.$$

Estimate the gap: $\widehat{\Delta}_{i,i^{(s)}}$.

Weak opponent statistic

Apply Range-Quantile within $A_r \setminus \{i\}$ and estimate

$$\widehat{Q}_{[1/8, 1/4]}((\Delta_{i,j})_{j \in A_r}).$$

This statistic detects rows containing many negative gaps.

Upper Bound on Fixed-confidence CWI (FC-CWI)

FC-CWI: at epoch k , compute $\hat{i}_k = \text{FB-CWI}(2^k)$, stop if either :

- The remaining arm $\hat{i}_k$ is certified to be i .
- The elimination frontiers are all certified to maintain active i^* .

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- Always better than $H_{\text{cw}}(\delta)$

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Theorem (Saad*, Thuot*, and Verzelen, 2026)

FC-CWI is δ -correct and, up to logarithmic factors, with probability at least $1 - \delta$:

$$N_\delta \lesssim H_{\text{cw}}(\delta) \wedge \min_s (H_{\text{certify}}(s, \delta) + H_{\text{explore}}(s, \delta)).$$

- Always better than $H_{\text{cw}}(\delta)$
- Balances exploration and certification

Upper Bound on Fixed-confidence CWI (FC-CWI)

FC-CWI: at epoch k , compute $\hat{i}_k = \text{FB-CWI}(2^k)$, stop if either :

- The remaining arm $\hat{i}_k$ is certified to be i .
- The elimination frontiers are all certified to maintain active i^* .

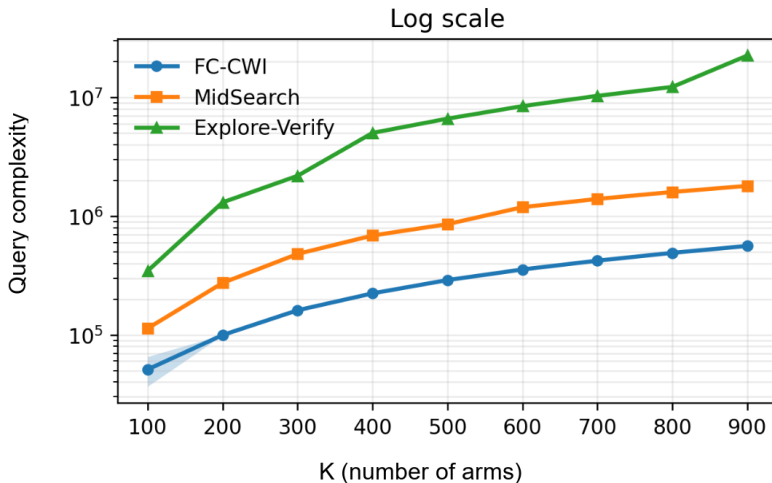
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- Achieves the best trade-off automatically in s_i

Numerical Experiment (Karnin, 2016; Maiti, Boczar, Jamieson, and Ratliff, 2024)



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- **Q4.** What are the advantages of bandit learning over classical (batch) learning?

Better adaptivity to the unknown structure.

Conclusion and Perspectives

Introduction to unsupervised active learning

Focus on Active Clustering

Focus on Condorcet Winner Identification

Conclusion and Perspectives

Summary

Ch.	Problem	Items	Features	Unknown Structure	Observation	Sub-Gaussian noise?	Fixed confidence? ¹	Non-asympt. optimal?
3	Clustering (Parametric)	$[n]$	$\mathbb{R}^d$	K clusters	One item All features	✓	✓	✓ local-minimax
4	Clustering (Non-parametric)	$[n]$	$\mathcal{X}$ ($d \leq \infty$)	K clusters	One item All features	✗ Any noise	✓	✗ No Lower Bound
5	Clustering (with feature selection)	$[n]$	$\mathbb{R}^d$	K clusters ($K = 2$)	One item One feature	✓	✓	~ $K = 2$, bilevel gap
6	Condorcet Winner Identification	$[K]$	$[K]$	Condorcet winner	Pairwise comparisons	✓ Bernoulli	✓	✓ local-minimax
7	Multiple Change Point Identification	$[0, 1]$	$\mathbb{R}$	Change-point locations (N among m)	Evaluate f at one point	✓	✓ ²	~ $N = m$

-
1. δ prescribed confidence
 2. and with η prescribed localization precision

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Algorithmic: detect → reconstruct → certify

Tools: concentration, sequential elimination, sub-sampling, tests

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Better computation efficiency.

Open Questions

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Merci pour votre attention

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